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SHOULDER MUSCLES

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I. INTRODUCTION

The muscles of the shoulder originate from the pectoral girdle of the upper limb and insert onto the humerus. These muscles are divided into four groups: anterior, posterior, medial and lateral, which form the walls of the axillary fossa one of the regions of the shoulder.

II. DESCRIPTIVE ANATOMY

A. ANTERIOR MUSCULAR GROUP

Composed of three muscles arranged in two layers:

a. Deep Layer

- 1. Subclavius Muscle**
 - Origin: First rib.
 - Body: Fusiform muscle.
 - Insertion: Subclavian groove on the inferior surface of the clavicle.
 - Action: Lowers the clavicle or raises the first rib.
 - Innervation: Subclavian nerve (collateral branch of the brachial plexus).
- 2. Pectoralis Minor Muscle**
 - Origin: 3rd, 4th, and 5th ribs.
 - Body: Flat, triangular muscle.
 - Insertion: Coracoid process.
 - Action: Lowers the coracoid process or elevates the ribs.
 - Innervation: Medial pectoral nerve (collateral branch of the brachial plexus).

b. Superficial Layer

- 3. Pectoralis Major Muscle**
 - Origin: Three heads:
 - Clavicular head: Medial two-thirds of the anterior border of the clavicle.

Sterno-costal head: Anterior surface of the sternum and the first six costal cartilages.

Abdominal head: Sheath of the rectus abdominis muscle.

- Body: Thick, triangular muscle.
- Insertion: Lateral lip of the intertubercular groove.
- Action: Adductor and medial rotator of the arm.
- Innervation: Medial and lateral pectoral nerves (collateral branches of the brachial plexus).

B. POSTERIOR MUSCULAR GROUP

1. Subscapularis Muscle

- Origin: Subscapular fossa.
- Body: Triangular muscle.
- Insertion: Lesser tubercle of the humerus.
- Action: Medial rotator of the arm.
- Innervation: Subscapular nerve (collateral branch of the brachial plexus).

2. Supraspinatus Muscle

- Origin: Medial two-thirds of the supraspinous fossa.
- Body: Thick, triangular muscle.
- Insertion: Superior facet of the greater tubercle of the humerus.
- Action: Lateral rotator and abductor of the arm.
- Innervation: Suprascapular nerve (collateral branch of the brachial plexus).

3. Infraspinatus Muscle

- Origin: Infraspinous fossa.
- Body: Broad, triangular muscle.
- Insertion: Posterior surface of the greater tubercle of the humerus.
- Action: Lateral rotator of the arm.
- Innervation: Suprascapular nerve (collateral branch of the brachial plexus).

4. Teres Minor Muscle

- Origin: Lateral border of the scapula.
- Body: Broad, flat muscle.
- Insertion: Inferior facet of the greater tubercle of the humerus.
- Action: Lateral rotator and accessory adductor of the arm.
- Innervation: Axillary nerve.

Note: The tendons of the subscapularis, supraspinatus, infraspinatus, and teres minor muscles form the rotator cuff, a frequent site of inflammation and injury.

5. Teres Major Muscle

- Origin: Lateral border of the scapula.
- Body: Flat muscle located below the teres minor.
- Insertion: Medial lip of the intertubercular groove.
- Action: Adductor and medial rotator of the arm.
- Innervation: Nerve to the teres major (collateral branch of the brachial plexus).

6. Latissimus Dorsi Muscle

- Origin: Medial iliac and sacral crests, spinous processes and interspinous ligaments, and inferior angle of the scapula.

- Body: Flat, triangular muscle.
- Insertion: Crest of the lesser tubercle of the humerus.
- Action: Adductor and medial rotator of the arm.
- Innervation: Thoracodorsal nerve (collateral branch of the brachial plexus).

C. MEDIAL MUSCULAR GROUP

1. Serratus Anterior Muscle

- Origin: First ten ribs.
- Body: Broad, flat muscle.
- Insertion: Medial border of the scapula.
- Action: Allows the scapula to glide against the thoracic wall.
- Innervation: Long thoracic nerve (collateral branch of the brachial plexus).

2. LATERAL MUSCULAR GROUP

1. Deltoid Muscle

- Origin: Lateral third of the clavicle, acromion, and spine of the scapula.
- Body: Thick, half-cone-shaped muscle covering the lateral part of the shoulder.
- Insertion: Lateral surface of the humerus at the deltoid tuberosity.
- Action: Abductor of the arm; also assists in flexion and extension.
- Innervation: Axillary nerve.

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